# SYSTEMIC AMINO ACID IDENTIFICATION AND SEPARATION

## CHROMATOGRAPHY:

Chromatography is a vital biophysical technique used to separate, identify, and purify components within a mixture, facilitating both qualitative and quantitative analysis. In protein purification, various chromatographic methods exploit distinct molecular characteristics to achieve effective separation.

**Ion Exchange Chromatography (IEC): Ion Exchange Chromatography (IEC)**

Ion exchange chromatography (IEC) is a widely used separation technique that relies on the reversible interaction between charged molecules and an oppositely charged stationary phase. It is commonly used for the purification of proteins, nucleic acids, and other biomolecules.

### Principle of Ion Exchange Chromatography

IEC is based on electrostatic interactions between charged molecules in a sample and oppositely charged functional groups on a resin (stationary phase). The molecules bind to the resin and are then eluted by increasing the ionic strength or changing the pH of the mobile phase.

**Types of Ion Exchange Chromatography**

1. **Cation Exchange Chromatography**
    - The stationary phase carries **negatively charged** groups.
    - It binds **positively charged** molecules (cations).
    - Example: Sulfonic acid ($-SO_3^-$) or carboxyl ($-COO^-$) functional groups.

2. **Anion Exchange Chromatography**
    - The stationary phase carries **positively charged** groups.
    - It binds **negatively charged** molecules (anions).
    - Example: Quaternary amine ($-N^+(CH_3)_3$) functional groups.

## Steps in Ion Exchange Chromatography

1. **Equilibration** – The column is equilibrated with a buffer at a specific pH to ensure the ion exchange resin is ready for binding.
2. **Sample Application** – The sample is loaded onto the column, and oppositely charged molecules bind to the resin.
3. **Washing** – Unbound molecules and weakly bound impurities are washed away.
4. **Elution** – The bound molecules are eluted by changing the buffer conditions (pH or salt concentration).
5. **Regeneration** – The column is cleaned and re-equilibrated for future use.

## Factors Affecting Ion Exchange Chromatography

1. **pH of Buffer** – Affects the charge of both the resin and the biomolecule, influencing binding and elution.
2. **Ionic Strength** – Increasing salt concentration (e.g., NaCl or KCl) competes with bound molecules, facilitating elution.
3. **Resin Type** – Different resins have varying charge densities and binding strengths.
4. **Flow Rate** – A slower flow rate improves resolution, while a faster flow rate speeds up separation.

## Applications of Ion Exchange Chromatography

- **Protein Purification** – Separating proteins based on charge differences.
- **Nucleic Acid Purification** – Isolating DNA and RNA.
- **Water Softening & Purification** – Removing unwanted ions (e.g., calcium, magnesium).
- **Pharmaceutical & Biotechnological Uses** – Purification of drugs, enzymes, and antibodies.

**Gel-Permeation(molecular sieve) chromatography** : Gel-permeation chromatography (GPC), also known as size-exclusion chromatography (SEC) or molecular sieve chromatography, is a technique used to separate molecules based on their size. This method is widely applied in polymer chemistry, biochemistry, and biotechnology for analyzing macromolecules such as proteins, polysaccharides, and synthetic polymers.

**Principle**

GPC relies on a porous stationary phase, typically made of polymeric or silica-based beads with controlled pore sizes. The separation process occurs due to **differential penetration** of molecules into these pores:

- **Larger molecules** are unable to enter the pores and pass through the column more quickly (eluting first).
- **Smaller molecules** diffuse into the pores, experiencing a longer retention time (eluting later).
- The separation is purely based on size and not on chemical interactions.

**Stationary and Mobile Phases**

- **Stationary phase:** Cross-linked polymer gels (e.g., polystyrene, dextran, agarose, silica) with defined pore sizes.
- **Mobile phase:** An appropriate solvent that carries the sample through the column (e.g., water, organic solvents like tetrahydrofuran).

**Applications**

1. **Polymer Analysis:** Determining molecular weight distribution and polydispersity of synthetic polymers.
2. **Protein Purification:** Separation of proteins based on size, useful in proteomics.
3. **Biomolecule Separation:** Purification of nucleic acids, polysaccharides, and other macromolecules.
4. **Removal of Small Impurities:** Desalting or buffer exchange in biochemical preparations.

**Advantages**

- Mild separation conditions (no strong chemical interactions).
- Suitable for sensitive biological molecules.
- Can determine molecular weight and size distribution.

**Limitations**

- Lower resolution compared to other chromatographic techniques.
- Requires careful selection of column material and solvent for accurate results.
- Limited applicability for molecules with similar sizes

## COLUMN CHROMATOGRAPHY

Combining these chromatographic techniques in a strategic sequence enhances the efficiency and specificity of protein purification processes. The choice of method depends on the unique properties of the target protein and the specific requirements of the purification task.

Column chromatography is a widely used technique for separating and purifying compounds based on their differential interactions with a stationary phase and a mobile phase. It is commonly used in chemistry, biochemistry, and pharmaceuticals.

### Principle

The technique relies on the different affinities of compounds for the stationary phase (solid material in the column) and the mobile phase (liquid or gas that moves through the column). Components of a mixture travel at different rates, leading to their separation.

### Components of Column Chromatography

1. **Column** – A cylindrical glass or plastic tube that holds the stationary phase.
2. **Stationary Phase** – A solid adsorbent material (e.g., silica gel or alumina) packed inside the column.
3. **Mobile Phase** – A liquid solvent or gas that moves the sample through the column.
4. **Sample** – The mixture of compounds to be separated.

5. **Eluent** – The solvent used to carry the sample through the column.
6. **Detector (optional)** – In some cases, detectors such as UV spectrophotometers are used to monitor separated compounds.

## Procedure

1. **Packing the Column** – The stationary phase (silica or alumina) is packed into the column with a solvent to form a uniform bed.
2. **Loading the Sample** – The mixture is dissolved in a small volume of solvent and applied to the top of the column.
3. **Elution** – The mobile phase is added to the column, allowing compounds to move at different speeds depending on their interaction with the stationary phase.
4. **Fraction Collection** – The separated compounds are collected in different fractions as they elute.
5. **Analysis** – The collected fractions can be analyzed using techniques such as TLC (thin-layer chromatography), spectroscopy, or mass spectrometry.

## Types of Column Chromatography

1. **Adsorption Chromatography** – Uses silica gel or alumina as a stationary phase, separating compounds based on polarity.
2. **Partition Chromatography** – Separation occurs due to differences in solubility between two liquid phases.
3. **Ion Exchange Chromatography** – Separates charged molecules using a stationary phase with charged groups.
4. **Size Exclusion Chromatography (Gel Filtration)** – Separates molecules based on size using porous beads.
5. **Affinity Chromatography** – Uses a stationary phase with specific binding interactions (e.g., enzyme-substrate, antigen-antibody).

## Applications

- **Purification of Compounds** – Used in the pharmaceutical and chemical industries.
- **Separation of Biomolecules** – Proteins, amino acids, and nucleotides are often separated using column chromatography.

- **Analysis of Complex Mixtures** – Used in forensic science, food chemistry, and environmental studies.
- **Preparation of Pure Substances** – Isolating natural products and synthetic chemicals.

**Affinity Chromatography** is a highly specific biochemical technique used to purify proteins, nucleic acids, or other biomolecules based on their specific interactions with a ligand. This method exploits the unique binding affinity between a target molecule and a complementary ligand that is immobilized on a stationary phase, typically a chromatographic resin or matrix.

**Principle of Affinity Chromatography**

1. **Ligand Attachment** – A specific ligand (e.g., antibody, enzyme substrate, or receptor) is covalently attached to a solid support (such as agarose, silica, or cellulose beads).
2. **Sample Loading** – A mixture containing the target molecule is passed through the column. The target binds selectively to the ligand, while unbound molecules are washed away.
3. **Washing** – The column is washed to remove non-specifically bound contaminants.
4. **Elution** – The target molecule is released from the ligand by changing conditions (e.g., altering pH, ionic strength, or using a competitive ligand).
5. **Recovery** – The purified target molecule is collected for further use.

**Applications of Affinity Chromatography**

- **Protein Purification** (e.g., purification of antibodies using Protein A or Protein G columns)
- **Enzyme Isolation** (e.g., immobilized metal affinity chromatography for His-tagged proteins)
- **Drug Development** (e.g., receptor-ligand studies)
- **Nucleic Acid Purification** (e.g., purification of RNA or DNA-binding proteins)
- **Biomarker Discovery** (e.g., isolating specific proteins from biological fluids)

**Advantages**

- High specificity and selectivity
- High purification yield with minimal steps

- Retains biological activity of purified molecules

**Disadvantages**

- Expensive ligands and matrix materials
- Potential ligand leakage, affecting purity
- Optimization required for efficient elution and recovery

**Paper Chromatography**

Paper chromatography is a laboratory technique used to separate mixtures of soluble substances. It is based on the movement of different components of the mixture through a **stationary phase (paper)** and a **mobile phase (solvent)**. The method is widely used for separating dyes, pigments, and chemicals like amino acids.

- **Principle of Paper Chromatography**

- The separation in paper chromatography is based on **capillary action** and the **differential solubility** of components in the solvent. The components of the mixture move at different rates due to their varying **affinity** toward the stationary phase (paper) and their **solubility** in the mobile phase (solvent).
- Each component travels a specific distance, which is expressed using the **Retention Factor (Rf value):**
- Rf=Distance traveled by the soluteDistance traveled by the solventRf = \ frac{\text{Distance traveled by the solute}}{\text{Distance traveled by the solvent}}Rf=Distance traveled by the solventDistance traveled by the solute

- Rf values help identify unknown substances by comparing them with known standards.

**Thin Layer Chromatography (TLC)**

Thin Layer Chromatography (TLC) is a simple and effective **separation technique** used to analyze mixtures based on their components' movement on a stationary phase under the influence of a mobile phase.

## Principle of TLC

TLC operates on the principle of **adsorption chromatography**. Different compounds travel at different rates on a TLC plate due to:

- **Affinity to the stationary phase** (silica gel, alumina, or cellulose).
- **Solubility in the mobile phase** (organic solvents or solvent mixtures).

## Materials Required

1. **TLC Plate** – Coated with a stationary phase (silica gel, alumina, or cellulose).
2. **Mobile Phase (Solvent System)** – A liquid solvent or solvent mixture.
3. **Sample** – The mixture to be analyzed.
4. **Capillary Tube or Micropipette** – To apply the sample.
5. **Developing Chamber** – A closed container to allow solvent movement.
6. **Visualization Method** – UV light, iodine chamber, or chemical reagents.

## TLC Procedure

### 1. Preparation of TLC Plate

- Take a **pre-coated TLC plate**.
- Draw a **light pencil line** (~1 cm from the bottom).

### 2. Sample Application

- Dissolve the sample in a volatile solvent.
- Spot the solution on the baseline using a capillary tube.
- Allow the spot to dry.

### 3. Development

- Place the plate in the **developing chamber** with the mobile phase.
- Ensure the solvent **does not touch the sample spot**.
- Let the solvent rise by capillary action.

**4. Visualization**

- Once the solvent reaches ~¾ of the plate, remove and dry it.
- Mark the **solvent front**.
- Observe under **UV light** or use **staining agents** (e.g., iodine, ninhydrin).

## Retention Factor (Rf) Calculation

Rf=Distance traveled by the compoundDistance traveled by the solventRf = \frac{\ text{Distance traveled by the compound}}{\text{Distance traveled by the solvent}}Rf=Distance traveled by the solventDistance traveled by the compound

**Interpretation:**

- **Higher Rf** → Less polar compound (moves farther).
- **Lower Rf** → More polar compound (moves less).

## Applications of TLC

**Identifying compounds**
**Checking purity of drugs**
**Monitoring chemical reactions**
**Forensic analysis (e.g., drug detection)**
**Food quality control**

**Gas Chromatography (GC)**

Gas Chromatography (GC) is a powerful **analytical technique** used to separate, identify, and quantify volatile compounds in a mixture. It is widely used in chemistry, pharmaceuticals, environmental analysis, and forensic science.

## Principle of Gas Chromatography

GC is based on the principle of **partition chromatography**. The sample is vaporized and carried by an **inert gas (carrier gas)** through a **stationary phase**

inside a column. Different components separate based on their **boiling points** and **interactions** with the stationary phase.

## Components of a Gas Chromatograph

1. **Carrier Gas (Mobile Phase)** – Typically **helium, nitrogen, or hydrogen**.
2. **Sample Injector** – Introduces the liquid or gaseous sample into the system.
3. **Column** – A long tube filled with the stationary phase (**capillary or packed column**).
4. **Oven** – Maintains and controls the temperature for separation.
5. **Detector** – Identifies and measures separated compounds (**FID, TCD, ECD, MS**).
6. **Data System** – Records and analyzes the chromatogram.

## Gas Chromatography Procedure

### 1. Sample Injection

- A small amount of sample is introduced into the injector.
- The sample is vaporized instantly.

### 2. Separation in the Column

- The vaporized sample is carried by the carrier gas through the column.
- Components separate based on their **boiling points** and **affinity to the stationary phase**.

### 3. Detection & Analysis

- A detector records the separated components as **peaks** in a chromatogram.
- The **retention time** (time taken by a compound to travel through the column) is used to identify compounds.

## Types of Gas Chromatography

1. **Gas-Solid Chromatography (GSC)** – Uses a solid adsorbent as the stationary phase.
2. **Gas-Liquid Chromatography (GLC)** – Uses a liquid-coated stationary phase (most common).

## Common GC Detectors

1. **Flame Ionization Detector (FID)** – Detects hydrocarbons and organic compounds.
2. **Thermal Conductivity Detector (TCD)** – Measures changes in heat conductivity.
3. **Electron Capture Detector (ECD)** – Detects halogenated compounds (e.g., pesticides).
4. **Mass Spectrometer (MS)** – Provides structural identification.

## Applications of Gas Chromatography

**Environmental Analysis** – Detecting pollutants and toxins.
**Pharmaceutical Industry** – Drug purity testing.
**Food Industry** – Identifying food additives and contaminants.
**Forensic Science** – Analyzing drugs, explosives, and poisons.
**Petrochemical Industry** – Fuel and oil composition analysis.

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## Advantages of Gas Chromatography

**High sensitivity and accuracy**
**Fast analysis of volatile compounds**
**Requires only small sample amounts**
**Can be coupled with mass spectrometry (GC-MS) for detailed analysis**

**Limitations**

**Only works for volatile and thermally stable compounds**
**Expensive equipment and maintenance**
**Requires skilled operation**

**Dye Ligand Chromatography**

Dye Ligand Chromatography is a type of **affinity chromatography** that uses immobilized **dye molecules** as ligands to selectively bind proteins, enzymes, and other biomolecules. It is widely used in **protein purification** and **enzyme isolation**.

## Principle of Dye Ligand Chromatography

- Synthetic dyes (e.g., **Cibacron Blue F3G-A**) mimic natural ligands like ATP, NADH, or coenzymes.
- These dyes are **covalently attached** to a stationary phase (usually agarose or sepharose beads).
- Proteins with complementary binding sites **interact with the dye** and get retained on the column.
- Weakly bound or unbound molecules are **washed away**, while bound proteins are **eluted** using a competitive agent (e.g., high salt, pH change, or specific ligands).

## Materials Required

1. **Dye Ligand Column** – A chromatography column with immobilized dye.
2. **Buffer Solutions** – Binding buffer, washing buffer, and elution buffer.
3. **Protein Sample** – The mixture containing target proteins.
4. **UV-Vis Spectrophotometer** – To analyze protein elution.

## Procedure for Dye Ligand Chromatography

### 1. Column Preparation

- Equilibrate the column with a **binding buffer** (e.g., phosphate buffer).

**2. Sample Loading**

- Apply the protein mixture to the column.
- Proteins with affinity for the dye will **bind** to the column.

**3. Washing**

- Unbound proteins are removed by washing with buffer.

**4. Elution**

- Bound proteins are **eluted** using:
  - High salt concentration (disrupts interactions).
  - pH change (alters protein charge).
  - Competitive ligands (e.g., NADH, ATP).

**5. Analysis**

- Collect and analyze the eluted fractions using **UV-Vis absorbance** or **SDS-PAGE**.

## Applications of Dye Ligand Chromatography

**Protein Purification** – Isolation of enzymes (e.g., dehydrogenases, kinases).
**Enzyme Purification** – Separation of ATP or NADH-dependent enzymes.
**Biomedical Research** – Studying protein-ligand interactions.

## Advantages of Dye Ligand Chromatography

**Cost-effective** compared to antibody-based affinity chromatography.
**Reversible binding** allows for easy purification.
**Broad specificity** for a variety of proteins and enzymes.

**Limitations**

**Non-specific binding** may require additional purification steps.
**Dye leakage** can affect downstream applications.

# Pseudoaffinity Chromatography

Pseudoaffinity Chromatography is a **protein purification technique** that mimics **affinity chromatography** but uses **synthetic ligands** instead of biological ones (e.g., antibodies, cofactors). These ligands interact with proteins through **non-specific but selective interactions**, such as **hydrogen bonding, ionic interactions, and van der Waals forces**.

## Principle of Pseudoaffinity Chromatography

- A **pseudoaffinity ligand** (e.g., dyes, metal ions, or synthetic peptides) is immobilized on a stationary phase.
- Proteins bind based on **mimicked biological interactions** rather than highly specific binding.
- **Washing** removes weakly bound or unbound proteins.
- **Elution** occurs using a change in **pH, ionic strength, or competitive binding agents**.

## Materials Required

1. **Pseudoaffinity Column** – Contains immobilized ligands (e.g., Cibacron Blue, metal ions).
2. **Buffer Solutions** –
    - **Binding buffer** (optimized for ligand-protein interaction).
    - **Elution buffer** (pH change, salt gradient, or competitive agent).
3. **Protein Sample** – The mixture to be purified.
4. **Spectrophotometer** – For analyzing protein fractions.

## Procedure for Pseudoaffinity Chromatography

### 1. Column Preparation

- Equilibrate the column with the **binding buffer**.

**2. Sample Application**

- Load the protein mixture onto the column.
- Proteins with affinity for the ligand **bind**, while others flow through.

**3. Washing**

- Remove non-specifically bound proteins with buffer washes.

**4. Elution**

- **Elute the target protein** by:
  - **pH shift** (disrupts interactions).
  - **Salt gradient** (changes ionic strength).
  - **Competitive ligand binding** (displaces protein from the column).

**5. Analysis**

- Collect fractions and analyze using **UV-Vis spectrophotometry or SDS-PAGE**.

## Examples of Pseudoaffinity Ligands

- **Dye Ligands** – Cibacron Blue for **enzyme purification** (e.g., kinases, dehydrogenases).
- **Metal Ions (IMAC)** – $Zn^{2+}$, $Ni^{2+}$ for **his-tagged protein purification**.
- **Synthetic Peptides** – Mimic biological receptors for **protein binding**.

## Applications of Pseudoaffinity Chromatography

**Enzyme purification** (e.g., dehydrogenases, kinases).
**Purification of nucleotide-binding proteins** (e.g., ATP-binding proteins).
**Separation of immunoglobulins and albumins**.
**Industrial-scale protein purification**.

## Advantages of Pseudoaffinity Chromatography

**Lower cost** than true affinity chromatography.
**Reusable ligands** (dyes, metals).
**Broader selectivity** while maintaining specificity.

**Limitations**

**Less specific than true affinity chromatography**.
**Non-specific binding may require additional purification steps**.
**Potential ligand leakage into the purified protein**.

**Fast and automated analysis**
**Works for a wide range of compounds**
**Quantitative and qualitative analysis**

**Limitations**

**Expensive equipment and maintenance**
**Requires trained personnel**
**Not suitable for highly volatile compounds (use GC instead)**

High-pressure chromatography, also known as **high-performance liquid chromatography (HPLC)**, is an advanced technique used for separating, identifying, and quantifying components in a mixture. It relies on high-pressure pumps to force a liquid solvent containing the sample through a column packed with a stationary phase, leading to efficient separation based on chemical properties.

## Key Features of High-Pressure Chromatography (HPLC)

1. **High Pressure (Typically 50–400 bar):** Improves the efficiency and speed of separation.
2. **Stationary Phase:** A packed column with small particle sizes (e.g., silica-based materials).
3. **Mobile Phase:** A liquid solvent (e.g., water, methanol, acetonitrile) that carries the sample.
4. **Detection Methods:** UV-Vis, fluorescence, refractive index, and mass spectrometry (LC-MS).
5. **Applications:**
    - Pharmaceutical analysis (drug purity, dosage)
    - Food safety testing
    - Environmental analysis (pollutant detection)
    - Biochemical research (protein and peptide analysis)

## Types of High-Pressure Chromatography

1. **Normal-Phase HPLC:** Separates nonpolar compounds using a polar stationary phase (e.g., silica) and a nonpolar solvent.
2. **Reverse-Phase HPLC (RP-HPLC):** The most common type, with a nonpolar stationary phase (e.g., C18 column) and a polar mobile phase.
3. **Ion-Exchange Chromatography:** Separates charged molecules based on ionic interactions.
4. **Size-Exclusion Chromatography (SEC):** Separates molecules based on size, used for protein and polymer analysis.

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